RA

Code No: 135SC

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, July/August - 2023
OPERATING SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

used paper July/Aug-2023

PART – A

(25 Marks)

1.a) What are operating system services? [2]
b) Write about user and OS interface. [3]
c) What is ready queue? [2]
d) What is the purpose of exec() system call? [3]
e) What is external fragmentation? [2]
f) If the main memory is 64KB and secondary memory is 128KB, how many pages and frames are there if page size is 2KB? How many entries are there in page table? [3]
g) What are various file access methods? [2]
h) What is the purpose of seek() function? [3]
i) What is hold-and-wait condition? [2]
j) What is resource allocation graph? [3]

PART – B

(50 Marks)

2. What is a system call? Explain different types of system calls. [10]

OR

3. Explain operating system role in process management and memory management. [10]

4. How can we use semaphores in process synchronization? Explain. [10]

OR

5. Perform preemptive and non-preemptive priority scheduling on following data. (5 – highest priority, 1 – lowest priority) and compute average waiting time: [10]

Process ID	Arrival time	Burst time	priority
P01	0	4	4
P02	1	5	1
P03	3	7	2
P04	5	1	5
P05	6	9	3

6.a) What is demand paging? What are the various steps involved when a page fault occurs? Explain.

b) What are the various methods to avoid thrashing? Explain them. [5+5]

OR

7. Explain how memory management is done through segmentation? [10]

8. Explain various disk scheduling algorithms with example. [10]

OR

9. What is a file? Explain various file allocation methods. [10]

10. Explain banker's algorithm for deadlock avoidance with an example. [10]

OR

11.a) What is access matrix? How is it implemented? Explain.

b) Explain how resource allocation graph can be used to deadlock avoidance. [5+5]

---ooOoo---

used paper July/Aug-2023